nature communications

Article | Article in Press

https://doi.org/10.1038/s41467-026-78514-9

A universal twist-slide coupling mechanism governs deformations of DNA, RNA, and RNA-DNA hybrid duplexes

Received: 18 February 2026

Accepted: 22 September 2026

Published online: 08 October 2026

Check for updates

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Nucleic acid duplexes, including DNA, RNA, and RNA–DNA hybrid (RDH), undergo structural deformations that are essential for genome replication, transcription, and RNA-mediated regulation. However, quantitative measurements and mechanistic understanding of these deformations remain limited. Here, we probe nucleic acid deformations by measuring helical twist changes using single-molecule magnetic tweezers, where minute twist variations (~0.01° per base pair) accumulate over kilobase-length duplexes and become experimentally detectable. Unexpectedly, we find that RDHs do not behave as a simple intermediate between DNA and RNA. Instead, RDHs exhibit markedly attenuated twist responses to salt and temperature, together with nearly vanishing twist–stretch coupling. Despite these distinct mechanical behaviors, all-atom molecular dynamics simulations reveal a common deformation pathway shared by all three duplex types: external perturbations first modulate base-pair slide, which is subsequently transduced into twist changes through a universal twist–slide coupling. Building on this mechanism, we derive approximate theoretical predictions that quantitatively reproduce the salt- and temperature-dependent twist responses of RDH. We further show that protein-induced RDH deformations also occur predominantly along the twist–slide coupling pathway. Together, our results establish twist–slide coupling as a universal framework governing nucleic acid deformations across duplex types and environmental conditions.

Helical twist is a fundamental structural and mechanical property of nucleic acid (NA) duplexes, governing their energetic landscape and biological function. In cells, twist is continuously generated, redistributed, and relaxed during essential processes such as DNA replication[1,2] transcription[3,4], chromatin remodeling[5,6], and gene editing[7–11]. Failure to properly manage twist leads to excessive torsional stress, supercoiling, and genome instability, necessitating the action of specialized enzymes including helicases and topoisomerases[12]. Beyond DNA, RNA duplexes and RNA-DNA hybrid (RDH) also occur in diverse cellular contexts, yet how their twist responds to environmental and mechanical perturbations remains incompletely understood.

RDHs occupy a particularly important but underexplored position in NA biology. They arise transiently during transcription when nascent RNA anneals to the DNA template, forming R-loops composed of RDH and a displaced single-stranded DNA (Fig. 1). R-loops are widespread in eukaryotic genomes, enriched at actively transcribed loci, and implicated in transcription regulation, DNA repair, chromosome segregation, and mitochondrial maintenance[13–20]. Aberrant accumulation of

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R-loops has been linked to genome instability and human diseases[21,22]. RDHs also play a central role in CRISPR-based genome editing, where guide RNA hybridizes with target DNA to form an R-loop that determines Cas protein binding, cleavage specificity, and reaction kinetics[23,24]. Despite their biological significance, the conformational and mechanical properties of RDH remain poorly characterized.

Recent experimental and computational studies have revealed that duplex twist is not a fixed geometric parameter, but changes with salt concentration, temperature, and mechanical stretching[25–28]. Surprisingly, DNA and RNA duplexes exhibit qualitatively different twist responses to the same external stimuli. Increasing stretching force overwinds DNA but unwinds RNA, while changes in salt and temperature induce distinct magnitudes in DNA and RNA[25,28–32]. These differences have been attributed to distinct duplex structures (B-form for DNA and A-form RNA in these studies) and their associated deformation pathways: twist-diameter coupling in DNA[25] and twist-groove coupling in RNA[28]. Together, these findings raise a fundamental question: how does the twist of RNA-DNA hybrids respond to environmental and mechanical cues, and what physical mechanism governs this response?

At first glance, one might expect RDH to behave as an intermediate between DNA and RNA duplexes[33,34]. However, the situation is more complex[35]. X-ray crystallography studies show that RDHs largely adopt A-form-like conformations[36–38], whereas NMR solution studies reveal that the DNA strand of RDH retains a mixture of A- and B-form conformations[34,39,40]. Given that duplex conformation critically shapes deformation pathways, whether RDHs follow the mechanisms established for DNA or RNA, or instead obey a distinct physical principle, remains an open question.

Here, we employ single-molecule magnetic-tweezers (MT) experiments and molecular dynamics (MD) simulations to systematically investigate twist changes in RNA-DNA hybrids induced by salt concentration, temperature, and stretching. Our results reveal that RDHs exhibit markedly attenuated twist responses to salt and temperature and an almost vanishing twist-stretch coupling, sharply contrasting with the pronounced and opposite twist-stretch behaviors of DNA and RNA[29–32]. Strikingly, despite these differences, we uncover a universal deformation mechanism shared by DNA, RNA, and RDH: external stimuli first induce changes in base-pair slide, which are then converted into twist variations through a common twist-slide coupling constrained by the sugar-phosphate backbone. The distinct twist responses of three duplex types arise not from different coupling mechanisms, but from how salt, temperature, and stretch reshape the slide parameter.

Measurements of twist changes using single-molecule magnetic tweezers (MT) provide a highly sensitive probe of subtle RDH deformations. Minute twist changes (~0.01° per base pair) accumulate over kilobase-length duplexes and generate experimentally detectable end rotations[25,28], enabling the detection of structural perturbations far smaller than the coordinate precision typically accessible by X-ray crystallography. Moreover, because MT measurements can be performed under in situ variations of salt concentration, temperature, and stretching force, they offer a realistic and controllable environment for studying nucleic acid deformations. Combined with all-atom molecular dynamics (MD) simulations, measurements of twist changes have provided important insights into deformation pathways in DNA and RNA duplexes[25,26,28] and are therefore well suited for investigating RDHs.

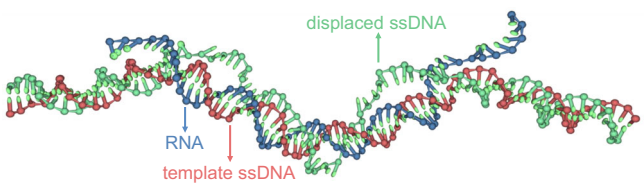


Fig. 1 | Illustration of RNA–DNA hybrid (RDH) formation during transcription. Nascent RNA anneals to the template DNA strand to form an RNA–DNA hybrid (RDH), leaving the non-template DNA strand displaced. The resulting three-stranded structure is termed an R-loop. The conformation shown is an illustrative snapshot from our coarse-grained oxNA simulations.

Understanding RDH deformations has important biological implications. In R-loops, the displaced single-stranded DNA (ssDNA) often wraps around the RDH to help preserve the linking number of the original DNA duplex. In such configurations, interactions between the displaced ssDNA and the RDH contribute substantially to the conformational energy of the R-loop, as demonstrated by recent statistical-mechanical models of R-loop formation[8]. Because RDH-ssDNA interactions depend sensitively on RDH geometry, deformations induced by salt, temperature, or mechanical force are expected to modulate R-loop stability and dynamics.

RDHs also serve as key intermediates in numerous biological processes, including CRISPR-mediated gene editing[24,41,42], reverse transcription[43–45], and RNA-DNA hybrid degradation[46–48], where they are tightly bound and reshaped by proteins. Protein binding inevitably induces RDH deformations and the associated deformation energies, which contribute to binding affinity, target recognition, and catalytic activity. Therefore, elucidating the intrinsic deformation pathways of RDHs is essential for understanding how proteins recognize, deform, and function on RNA-DNA hybrids.

Results

Experiments of twist changes

In MT experiments, two ends of an RDH duplex were attached to a glass slide and a magnetic bead, respectively (Fig. 2A). Both ends had multiple attachment points to ensure the RDH duplex was torsionally constrained. In each experiment, we held one RDH duplex at a constant stretching force (0.3 pN), changed salt concentration or temperature, and then measured the rotation of the magnetic bead, which corresponds to RDH twist change. See more measurement details in Supplementary Fig. 1 and Supplementary Note 1, and our previous studies[25,28]. The experimental measurement of twist change can achieve a high precision, because a tiny twist change, e.g., 0.01°/bp, can accumulate over the long duplex, 13 kbp, and cause the end rotation of 130°, which is detectable. For the same RDH molecule, we measured the twist change for a series of salt concentrations, c_{salt}, and obtained the results in Fig. 2E–H.

The RDH twist increases steadily with both NaCl and KCl concentrations (solid lines in Fig. 2E, G). We obtained similar results for other monovalent ions: LiCl, RbCl, and CsCl (see Supplementary Fig. 2), which suggests that the salt effect on RDH twist is not ion-specific and is probably caused by electrostatic screening instead of ion-specific binding on DNA.

For comparison, we show the twist changes of DNA duplex and RNA duplex with the same sequences (Fig. 2E, G), where the experimental results for DNA and RNA have been measured and reported in our previous studies[25,28]. The magnitude of salt-induced twist change over the concentration range (0.01–1 M) follows the order for different duplex types: RDH ≤ DNA < RNA, which does not follow the expectation that properties of RDH lie between those of DNA and RNA, or are biased toward A-form structure[33,34]. Specifically, when the NaCl or KCl concentration increases from 0.01 to 1 M, the increase in twist angle of RNA is twice that of twist change for both DNA and RDH when NaCl or KCl concentrations increase from 0.01 to 1 M.

We further examined the effects of divalent cations ($MgCl_2$, $CaCl_2$, and $MnCl_2$) on duplex twist. Increasing divalent-cation concentration overwinds all three duplex types (Supplementary Fig. 3), with RNA showing the strongest response and DNA and RDH exhibiting similarly weaker responses. Therefore, the relative behavior among DNA, RNA,

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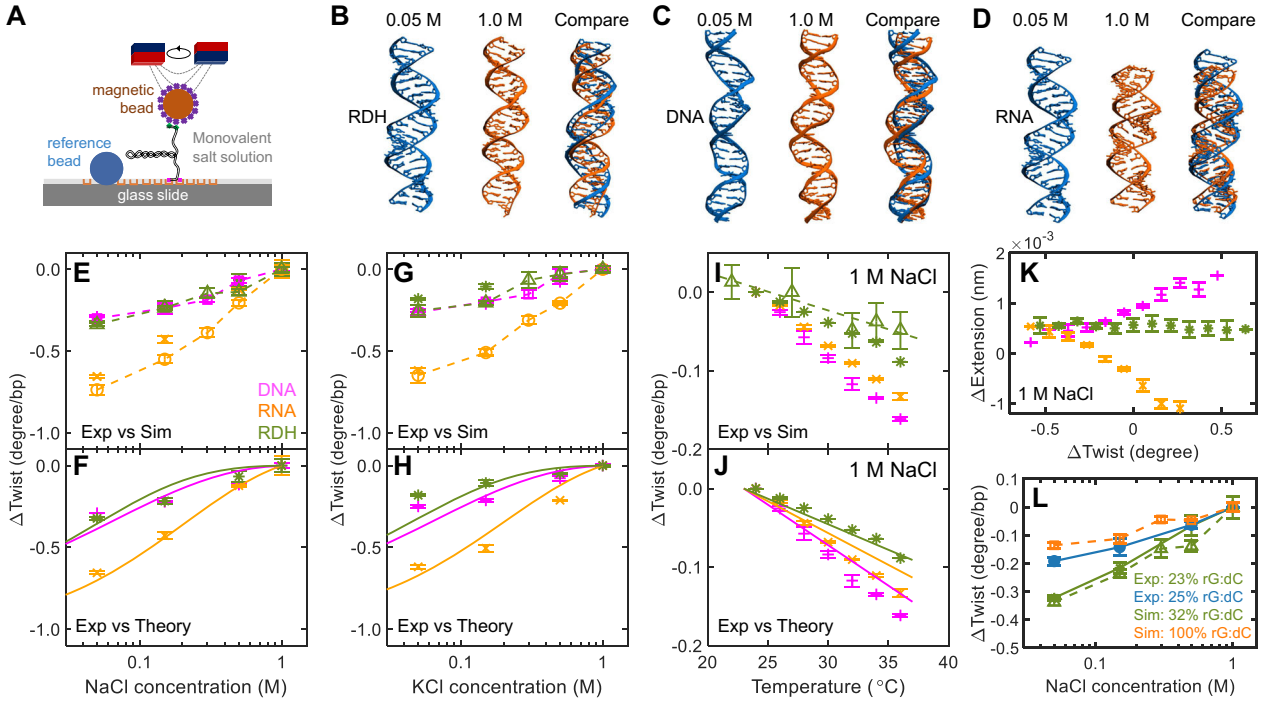


Fig. 2 | Comparison of DNA, RNA, and RDH deformation from MT measurements and MD simulations. A The setup of the MT experiment. **B**–**D** Snapshot of MD simulations showing duplex deformations induced by the variation of NaCl concentration, adapted from the averaged conformations of production trajectories after equilibrium. **E**, **F** Twist change as a function of NaCl concentration from MT measurements (symbols +, x, and * for DNA, RNA, and RDH, respectively), MD simulations (open symbols and dashed lines), and theoretical calculations (solid lines) for DNA, RNA, and RDH duplexes, respectively. **G**, **H** The same as in (**E**, **F**) except in KCl solution. **I**, **J** Twist change as a function of temperature in 1 M NaCl from MT experiments (symbols +, x, and * for DNA, RNA, and RDH, respectively), MD simulations (open symbol for RDH), and theoretical predictions (solid lines). Linear fits to experimental data yield the slopes of −0.0136 ± 0.0010 deg/°C for DNA, −0.0113 ± 0.0006 deg/°C for RNA, and −0.0071 ± 0.0009 deg/°C for RDH. The theoretical calculations predict the slopes of −0.0103, −0.0081, and −0.0065 deg/°C for DNA, RNA, and RDH duplexes, respectively. **K** Extension versus twist change in 1 M NaCl at 8 pN from MT experiments for DNA, RNA, and RDH duplexes, respectively. **L** Twist changes for different sequences from experiments and simulations. The twist changes for DNA and RNA with salt and temperature are replicated from our previous studies[25,28]. The averages and standard errors from five molecules are plotted as data points and error bars. The mean value over the corresponding MD trajectory block-averaged with a 100-ns interval, and standard errors are plotted as data points and error bars. Source data are provided as a Source Data file.

and RDH is preserved across ionic conditions, with RDH displaying an attenuated twist response compared with RNA.

We also measured RDH twist change for a series of temperatures, T, using a similar experimental procedure: holding one RDH molecule by MT, changing the temperature, and then measuring the rotation of the magnetic bead. Figure 2I shows that the twist angle decreases with the increase of T at 1 M NaCl. Linear fits to the data lead to temperature-dependent twist change as −0.0136 ± 0.0010, −0.0113 ± 0.0006, and −0.0071 ± 0.0009 deg/°C for DNA, RNA, and RDH, respectively. The temperature-dependent twist changes in 0.05 M and 0.15 M NaCl are also shown in Supplementary Fig. 4. Interestingly, the temperature-induced twist change also follows the trend between three duplexes: RDH < DNA < RNA.

To measure the interplay of RDH twist with stretching, we used a slightly different approach. Instead of stretching RDH and then measuring RDH twisting, we twist RDH and then measure RDH extension. Supplementary Fig. 5 shows a representative process where we rotated RDH positively and recorded the extension at 1 M NaCl. As shown by Fig. 2K, RDH extension is insensitive to twist change, and that is why one cannot change RDH twist by stretching RDH. In Fig. 2K, the variation of RDH extension is normalized by the number of base pairs: $\Delta L_{\mathrm{ext}}^{\mathrm{bp}} = \Delta L / N_{\mathrm{bp}}$. DNA duplex and RNA duplex have opposite twist-stretch coupling, while RDH has almost no twist-stretch coupling, seeming to be an average of DNA and RNA.

To investigate sequence effects, we measured salt-induced twist changes using a different RDH sequence (Fig. 2L). Most of the experimental results presented in this study, including those shown in Fig. 2E–K, were obtained using a 13.6-kbp RDH containing 23% rG:dC base pairs (where r denotes the RNA strand and d denotes the DNA strand). The second RDH construct has the length of 11.4-kbp but a slightly higher rG:dC content of 25%. Although the overall trend remains unchanged, the second sequence exhibits a weaker twist response to changes in salt concentration. All sequences studied are described in Supplementary Note 17.

Simulations of twist changes

We performed all-atom MD simulations to investigate the molecular mechanisms of twist changes induced by salt, temperature, and stretching (Fig. 2B–D). We analyzed the simulation results of DNA and RNA from our earlier work[25,28]. First, our MD simulations of 25-bp NA duplexes almost reproduced salt-induced twist changes for three duplexes and the order: RDH ≤ DNA < RNA (Fig. 2E, G). Second, our simulations reproduced temperature-induced twist changes semi-quantitatively (Fig. 2I). Third, our simulations reproduced the opposite twist-stretching couplings of DNA and RNA, and nearly zero twist-stretching coupling of RDH (Supplementary Fig. 6).

Finally, simulations of a 25-bp RDH composed entirely of rG:dC base pairs exhibited a weaker twist response to salt. This sequence dependence is quantitatively consistent with experimental observations (Fig. 2L). We specifically examined the effect of rG:dC content because rG:dC is the most thermodynamically stable among the four base-pair combinations. A higher rG:dC content is therefore expected to reduce the conformational flexibility of RDH and consequently attenuate its twist response. Additional results concerning the role of

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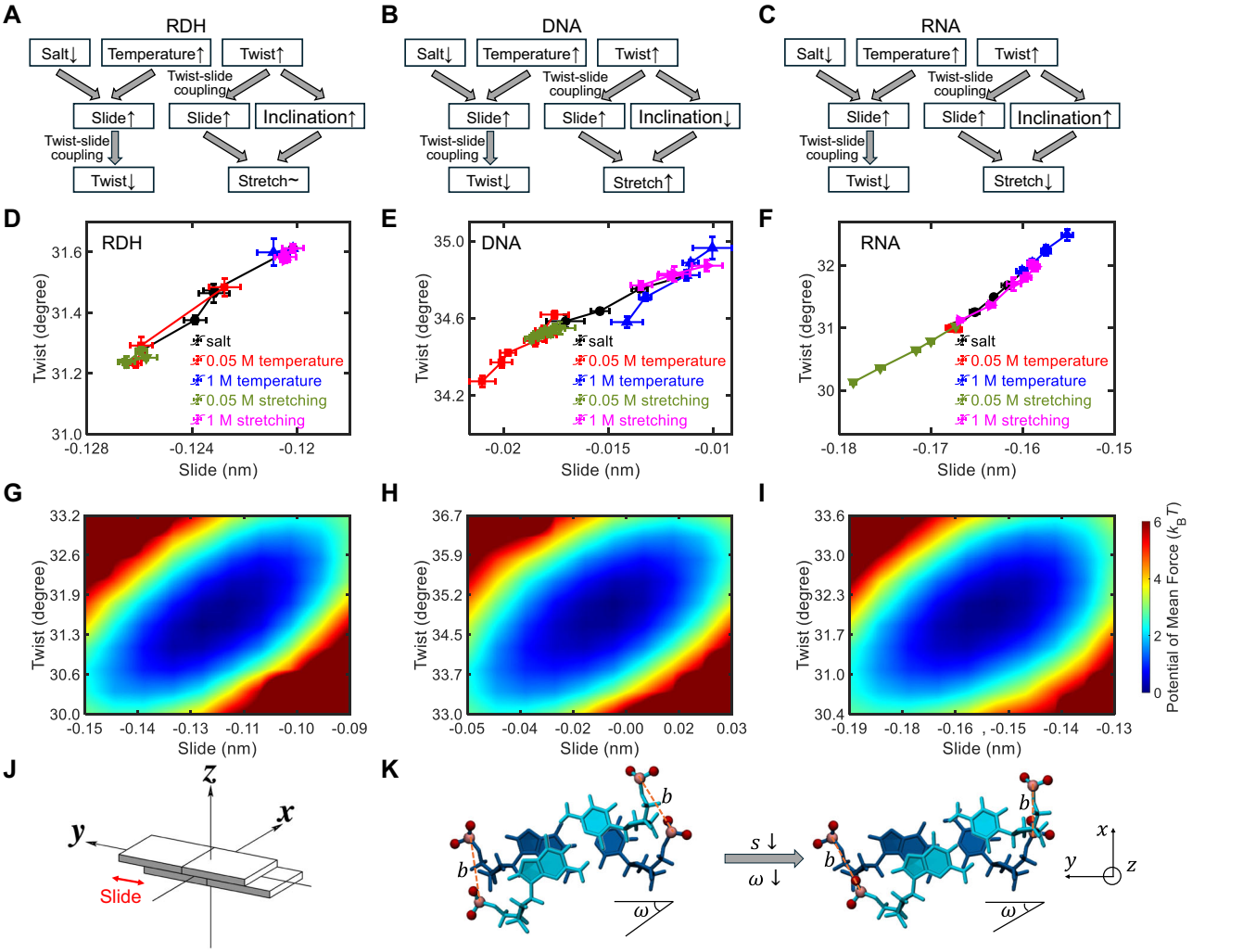


Fig. 3 | Deformation pathways in RDH, DNA, and RNA duplexes. A–C Illustrations of the twist changes in RDH, DNA, and RNA duplexes by different stimuli mediated by twist-slide coupling. **D–F** Twist versus slide with varying salt, temperature, and force for RDH, DNA, and RNA duplexes from simulations. The twist and slide values for each data point were calculated by averaging over all base pairs (excluding the terminal base pairs) and all sampled conformations. **G–I** The two-dimensional potential of mean force (PMF) with respect to twist and slide for RDH, DNA, and RNA obtained from simulations in 0.05 M NaCl. **J** Illustrations of the slide parameter, adapted from 3DNA[50]. **K** Twist-slide coupling restrained by sugar-phosphate backbone length, *b*. Each data point represents the mean value over the corresponding MD trajectory block-averaged with a 100-ns interval, and standard errors are plotted as data points and error bars. Source data are provided as a Source Data file.

rG:dC content are presented in the following part and Supplementary Table 1: a higher rG:dC content appears to have a larger twist rigidity, which may cause weaker twist responses.

To assess whether our simulations were sufficiently equilibrated, we performed convergence diagnostics. As shown in Supplementary Figs. 7–9, backbone RMSD, twist, and slide reach the equilibrium very quickly and are all in stable distributions within the analyzed production window. Block-averaged values from different trajectory segments were consistent within statistical uncertainty.

Because RDHs in vivo often contain single-stranded flanking regions, we simulated a 25-bp RDH duplex with 6-nt overhangs at both ends. The salt-dependent twist changes remained qualitatively unchanged (Supplementary Fig. 10), with only minor quantitative differences, likely arising from interactions between the flanking and duplex regions. Such effects may cause length-dependent deformations for short RDHs but should be negligible for kilobase-length duplexes. We also performed oxNA[49] simulations of 60-bp RDH to validate the length dependence. As shown in Supplementary Fig. 11, the simulation results reproduce both the salt- and temperature-induced twist change quantitatively.

Twist-slide coupling as a common deformation pathway

Analysis of our simulation results surprisingly uncovers a common universal mechanism for RDH, DNA and RNA twist changes: salt, temperature, or stretching first induces the slide change, and then the slide change is converted into the twist change via the twist-slide coupling (Fig. 3).

Slide is a duplex structural parameter to describe the displacement of two consecutive base pairs along the direction perpendicular to the base-pairing direction (Fig. 3J). Our analysis shows that the twist-slide coupling is common for all three NAs and all three stimuli. We also find a molecular mechanism for the twist-slide coupling: the restraint of the backbone length between two consecutive base pairs. As shown in Fig. 3K, two consecutive base pairs in a duplex are connected through the backbone of P-O5′-C5′-C4′-C3′-O3′-P, with approximately fixed contour length (*b*) between consecutive phosphate regions. The movement of the upper base pair with respect to the negative y-axis (the slide chain) tends to vary the backbone length, and such variation can be relieved by the decrease of the twist angle.

Figure 3D–F shows the scatter plots of twist and slide values for RDH, DNA, and RNA under varying salt concentrations, temperatures, and stretching. Slide was calculated and plotted using its signed value

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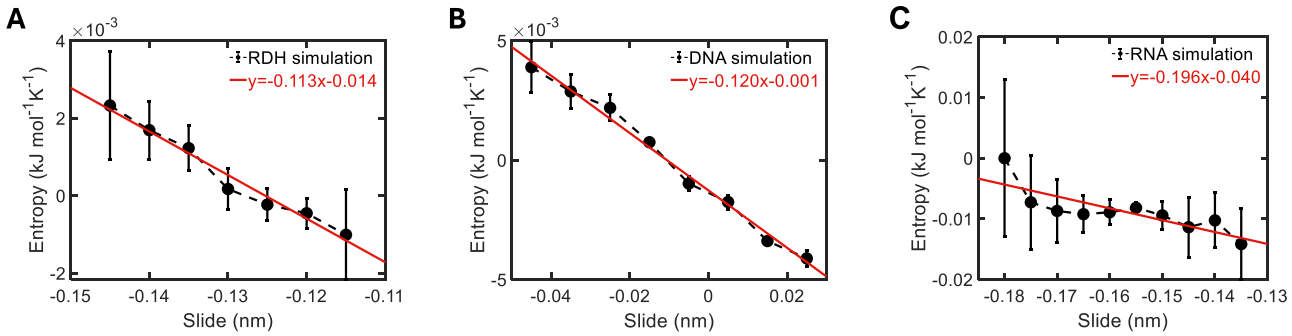


Fig. 4 | Dependence of conformational entropy on the slide extracted from simulations in 1 M NaCl for (A) RDH, (B) DNA, and (C) RNA, respectively. The mean value over the corresponding MD trajectory block-averaged with a 100-ns interval, and standard errors are plotted as data points and error bars. Source data are provided as a Source Data file.

following the standard 3DNA convention[50]. In three duplexes, stimulus-induced changes in twist remain strongly correlated with changes in signed slide. The data points approximately collapse onto a master curve, supporting the prevalence of twist-slide coupling in these three duplexes.

The twist-slide coupling can be extracted from the two-dimensional potential of mean force (PMF) over twist and slide:

$$P_{\mathrm{sim}}(\omega, s) = -\left(k_{\mathrm{B}}T\right)\ln[\Omega(\omega, s)], \tag{1}$$

where k_{B} is the Boltzmann constant, $\Omega(\omega, s)$ is the relative density of NA duplex conformations at a given ω and s (see calculation details in Supplementary Note 8). Figure 3G–I shows the PMF at 0.05 M NaCl for DNA, RNA, and RDH, respectively. Fitting the PMF with the following equation yields the twist-slide coupling constant, $k_{\omega G}^{\mathrm{bp}}$:

$$P^{\mathrm{bp}} \approx \frac{1}{2}k_{\omega}^{\mathrm{bp}}(\Delta\omega)^2 + \frac{1}{2}k_{s}^{\mathrm{bp}}(\Delta s)^2 + k_{\omega s}^{\mathrm{bp}}\Delta\omega\Delta s, \tag{2}$$

where here k_{ω}^{bp} and k_{s}^{bp} are the rigidity of twist and slide, respectively, $\Delta\omega$ and Δs are deviations of twist and slide from their global minimum. The superscripts of bp represent that these coefficients are normalized for one base pair. The fitting quantity for RDH at 0.05 NaCl is $k_{\omega}^{\mathrm{bp}} = 0.22 \pm 0.02\, k_{\mathrm{B}}T/\mathrm{deg}^2$, $k_{s}^{\mathrm{bp}} = 555 \pm 48\ \ k_{\mathrm{B}}T/\mathrm{nm}^2$, $k_{\omega s}^{\mathrm{bp}} = -8.52 \pm 0.94\, k_{\mathrm{B}}T/(\mathrm{deg}\cdot\mathrm{nm})$. The fitting values of k_{ω}^{bp}, k_{s}^{bp}, and $k_{\omega s}^{\mathrm{bp}}$ for other c_{salt} and other monovalent ion species can be found in Supplementary Table 2. The k_{ω}^{bp}, k_{s}^{bp}, and absolute value of $k_{\omega s}^{\mathrm{bp}}$ follow the order of RNA > RDH > DNA.

Theoretical calculation of salt-induced twist changes

To rationalize the experimentally observed salt-dependent twist changes, we developed an approximate theoretical model. Lowering the salt concentration enhances electrostatic repulsions between negatively charged phosphate (P) groups. These repulsions tend to increase both the inter-strand P–P distance, associated with the duplex diameter, D, and the cross-major-groove P–P distance, associated with the major-groove width, G.

To quantify these effects, we calculated the electrostatic interaction energy, $E(G, D)$, using the Manning formalism[51]. The corresponding electrostatic forces that promote increases in D and G are $f_D = -\partial E/\partial D$ and $f_G = -\partial E/\partial G$, respectively. The electrostatic force acting on slide can then be obtained using the chain rule:

$$f_s = -\frac{\partial E}{\partial s} \approx -\frac{\partial E}{\partial G}\frac{\partial G}{\partial s} - \frac{\partial E}{\partial D}\frac{\partial D}{\partial s}. \tag{3}$$

Here, $\partial G/\partial s$ and $\partial D/\partial s$ were obtained from analysis of MD simulation trajectories. As shown in Supplementary Fig. 12, $\partial G/\partial s = -2.22 \pm 0.11$, -4.50 ± 0.16, -4.35 ± 0.19 for DNA, RNA, and RDH duplexes, respectively, in 0.05 M NaCl. And $\partial D/\partial s = -0.799 \pm 0.004$, -0.204 ± 0.009, -0.594 ± 0.010 for DNA, RNA, and RDH duplexes in 0.05 M NaCl.

The force on the slide subsequently induces twist changes through twist–slide coupling. The deformation energy per base pair is written as

$$P^{\mathrm{bp}}(c_{\mathrm{salt}}) \approx \frac{1}{2}k_{\omega}^{\mathrm{bp}}(\Delta\omega)^2 + \frac{1}{2}k_{s}^{\mathrm{bp}}(\Delta s)^2 + k_{\omega s}^{\mathrm{bp}}\Delta\omega\Delta s - \Delta f_s\Delta s, \tag{4}$$

where $\Delta f_s = f_s(c_{\mathrm{salt}}) - f_s(1\mathrm{M})$ is the salt-induced change in the electrostatic force on slide relative to the 1 M salt condition. Minimization of P^{bp} with respect to both $\Delta\omega$ and Δs gives

$$\Delta\omega(c_{\mathrm{salt}}) \approx -\frac{k_{\omega s}^{\mathrm{bp}}}{k_{\omega}^{\mathrm{bp}}k_{s}^{\mathrm{bp}} - \left(k_{\omega s}^{\mathrm{bp}}\right)^2}\Delta f_s. \tag{5}$$

Using the Manning formalism to calculate Δf_s and substituting the resulting values into Eq. (5), we obtained the predicted salt-dependent twist changes shown by the solid lines in Fig. 2F, H, which agree reasonably well with the experimental measurements.

The model also captures a substantially weaker salt response of RDH than RNA. RDH possesses a wider major groove than RNA. Consequently, in the theoretical calculation, electrostatic repulsion across the groove f_G is smaller in RDH than in RNA (Supplementary Fig. 13), which leads to a weaker salt dependence of RDH twist compared with RNA. See more details in Supplementary Note 10.

Theoretical calculation of temperature-induced twist changes

To understand the temperature dependence of duplex twist, we developed an approximate theoretical model based on the temperature dependence of the slide coordinate. The conformational free energy associated with slide, s, can be decomposed into energetic and entropic contributions: $F(s) = U(s) - TS(s)$, where $U(s)$ and $S(s)$ denote the effective potential energy and conformational entropy, respectively. Increasing temperature enhances the entropic contribution, thereby shifting the free-energy landscape and altering the equilibrium distribution of slide.

To determine the entropy as a function of slide, $S(s)$, we first obtained the slide probability distributions, $\Omega(s)$, from MD simulations and calculated the corresponding potential of mean force (PMF): $F(s) = -k_{\mathrm{B}}T\ln\Omega(s)$. Assuming that the effective energetic term $U(s)$ is approximately temperature-independent over the investigated temperature range, subtracting PMFs obtained at different temperatures (Supplementary Fig. 14) largely eliminates the energetic contribution and allows extraction of the entropy profile $S(s)$ (Fig. 4). Details of this

procedure are provided in Supplementary Note 11. Similar approaches have been employed in previous studies[25,28].

We then define the entropy gradient with respect to slide as $k_{\text{entropy}} = \partial S/\partial s$. When the temperature changes from a reference temperature T_0 by $\Delta T = T - T_0$, the entropy gradient generates an effective entropic force driving slide deformation: $\Delta f_T = k_{\text{entropy}} \Delta T$. Following the same framework as in Eq. (5), this effective force is transduced into the twist change through twist–slide coupling:

$$\Delta\omega(\Delta T) \approx \frac{-k_{\omega s}^{\text{bp}}}{k_{\omega}^{\text{bp}} k_{s}^{\text{bp}} - \left(k_{\omega s}^{\text{bp}}\right)^2} \Delta f_T. \tag{6}$$

Accordingly, the slope of temperature-induced twist change is predicted to be

$$k_T^{\text{bp}} = \frac{\Delta\omega}{\Delta T} \approx \frac{-k_{\omega s}^{\text{bp}}}{k_{\omega}^{\text{bp}} k_{s}^{\text{bp}} - \left(k_{\omega s}^{\text{bp}}\right)^2} k_{\text{entropy}}. \tag{7}$$

Using this model, we obtained $k_T^{\text{bp}} \approx -0.0103\,\text{deg}/^\circ\text{C}$ for DNA, $-0.0081\,\text{deg}/^\circ\text{C}$ for RNA, and $-0.0065\,\text{deg}/^\circ\text{C}$ for RDH in 1 M NaCl, in reasonable agreement with the experimental measurements.

It should be emphasized that this theoretical calculation model is phenomenological and relies on the temperature dependence of the slide distribution. In reality, temperature affects numerous components of the nucleic-acid system, including solvent and ion interactions, as well as many local structural parameters and intramolecular interactions. These effects may influence slide through multiple microscopic pathways. Therefore, the observed temperature dependence of the slide should be viewed as the cumulative result of many temperature-dependent processes.

Nevertheless, our results highlight slide as a key structural parameter mediating temperature-induced twist changes. The twist–slide coupling remains robust across different duplex types and temperatures, and the above theoretical framework, based solely on slide redistribution and twist–slide coupling, quantitatively reproduces the experimentally observed temperature-dependent twist responses.

Origin of the vanishing twist–stretch coupling in RDH

We next explain why twisting an RDH produces almost no change in its extension. Previous studies have shown that the helical twist of a nucleic acid duplex can be expressed as helical twist, $\omega = \omega_{\text{local}}/\cos\theta$, where ω_{local} is the local twist angle about an axis perpendicular to the base pairs, and θ is the inclination angle between this local axis and the helical axis. Accordingly, an increase in helical twist can arise from increases in either ω_{local} or θ (Supplementary Fig. 15). Owing to twist–slide coupling, twisting RDH is also accompanied by a change in slide.

As shown in previous studies[31,32,52], the duplex extension is determined by the helical rise (H-rise), which is related to the local structural parameters through H-rise = local rise × cos(θ) + slide × sin(θ). Consequently, twisting RDH increases both the inclination angle and slide, which causes the decrease of local rise × cos(θ) and the increase of slide × sin(θ). These two contributions largely compensate for each other, resulting in an almost unchanged H-rise and hence a nearly vanishing twist–stretch coupling (Supplementary Fig. 15). In contrast, the relative contributions of the two terms differ in DNA and RNA, giving rise to opposite twist–stretch responses: overwinding elongates DNA but shortens RNA. We also extracted the twist-stretch Pearson correlation coefficients from simulation results. As given in Supplementary Table 4, the twist-stretch correlation coefficients are within 0.16 ~ 0.20 for DNA. This is in stark contrast to RNA, which gives the value of −0.40 ~ −0.29 for twist-stretch correlation coefficients. Lastly, the twist-stretch correlation coefficient is weakly positive for RDH, with the value of 0.03 ~ 0.08. Our results of twist-stretch coupling extracted from simulations (Supplementary Table 5) are in good agreement with previous experiments and simulations[30–33,52].

Structural parameters mediating RDH deformations

We next examine the structural parameters that mediate RDH deformations. Because RNA duplexes typically adopt the A-form and DNA duplexes typically adopt the B-form, an important question is which conformational state the DNA strand adopts in RDHs and how this state influences RDH deformation pathways.

Recently, Krepl and co-workers performed comprehensive MD simulations of RDHs using different force fields[35] and characterized their conformational states through sugar pucker and backbone dihedral angles (Fig. 5A). Following a similar strategy, we analyzed RDH conformations using two representative force fields (Fig. 5B). The first is the OL15/OL3 combination for DNA/RNA[53,54], belonging to the AMBER force-field family and used for the simulations presented in Figs. 2–4. The second is CHARMM36, which was previously shown by Krepl and co-workers to better reproduce certain structural features of RDHs[35]. Although our simulations employed sequences different from those used in the previous study[35], they reveal similar trends: (i) the RNA strand predominantly adopts the A-form in both force fields; (ii) the DNA strand mainly adopts the B-form in OL15/OL3; and (iii) the DNA strand exhibits a substantially higher A-form population in CHARMM36. The conformational state of the DNA strand is reflected in the distributions of the δ and χ dihedral angles as well as the sugar pucker. In general, the first peak in each distribution shown in Fig. 5B corresponds to the A-form, whereas the second peak corresponds to the B-form. For sugar puckering, the two peaks are associated with the C3′-endo (~10°) and C2′-endo (~150°) conformations, respectively. In addition, the two force fields produce different distributions of other structural parameters, including inclination, buckle, and X-displacement (Fig. 5C).

Despite these differences, we find that the δ angle serves as a key structural parameter mediating RDH deformations because it correlates strongly with both slide and twist (Fig. 5D–E and G–H). These correlations are observed in both force fields, despite their distinct A/B-form preferences. This observation suggests that environmentally induced changes in slide and twist are realized, at least in part, through rotations of the δ angle.

Indeed, simulations performed at different NaCl concentrations show systematic changes in δ (Fig. 5F, I). Interestingly, the two force fields predict different strand-specific responses. In OL15/OL3, both the RNA and DNA strands exhibit significant δ rotations, whereas in CHARMM36, a significant δ rotation is observed primarily in the DNA strand. This difference may arise from the distinct correlation strengths between δ and slide. In CHARMM36, δ in the DNA strand is strongly correlated with slide, with a correlation coefficient of 0.72, whereas δ in the RNA strand shows only a moderate correlation (correlation coefficient = 0.32).

Importantly, despite these force-field-dependent differences at the microscopic level, the predicted salt-dependent twist changes are highly similar in the two force fields and are both consistent with experimental measurements (Fig. 5J). Furthermore, the effective PMF generated by twist–slide coupling in CHARMM36 (Fig. 5K) closely resembles that obtained using OL15/OL3 (Fig. 3G–I). Together, these results demonstrate that twist–slide coupling is robust across different force fields and constitutes a major driving mechanism underlying RDH twist changes.

We also examine the correlations between twist, slide, and other structural parameters, including diameter, major groove width, inclination, x-displacement, torsion angles (α, β, γ, ε, and ζ), and sugar pucker for DNA, RNA, and RDH duplexes. The correlation between twist and some geometrical parameters appears to be strong but is weaker than twist–slide coupling (Supplementary Figs. 16 and S17).

https://doi.org/10.1038/s41467-026-78514-9

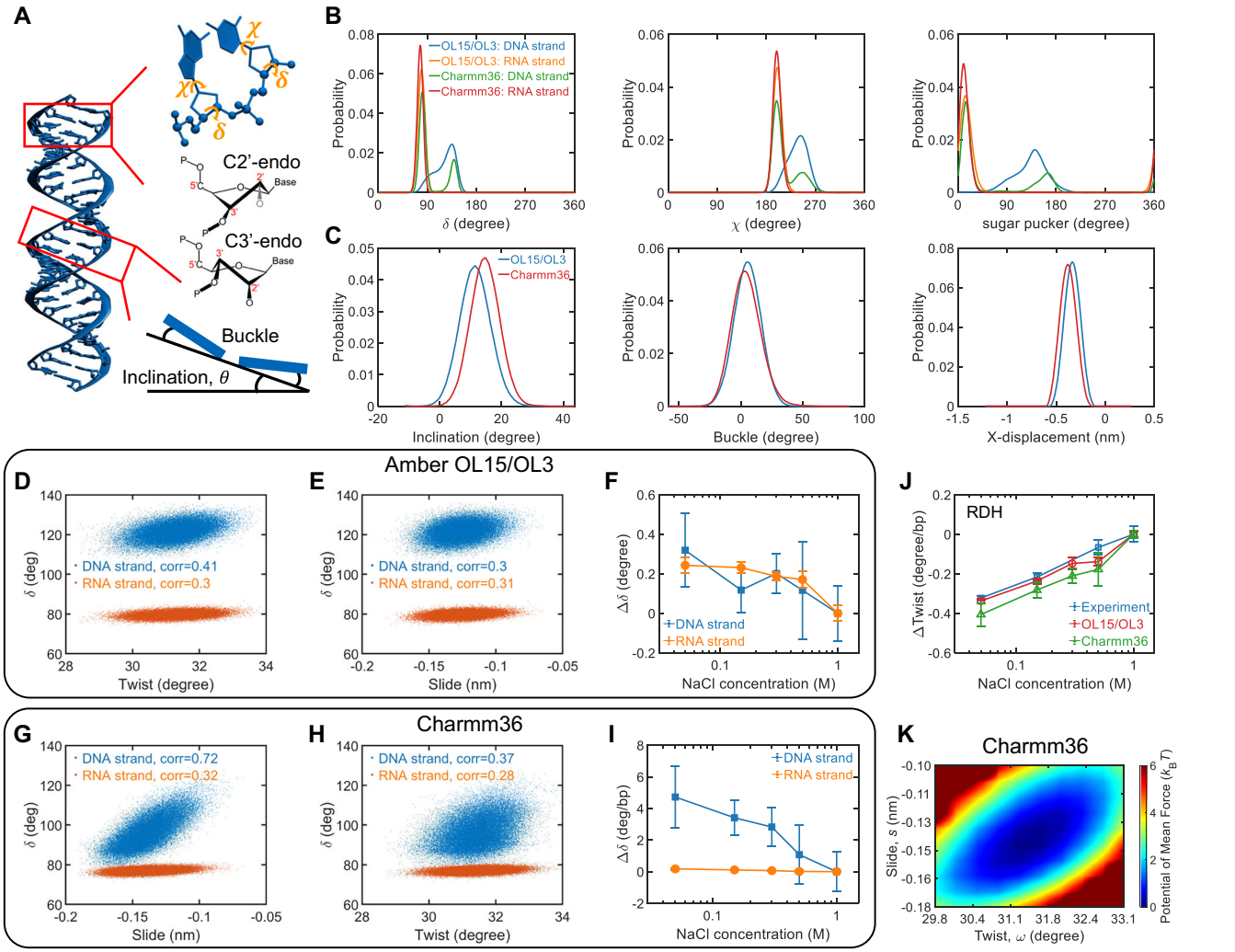


Fig. 5 | Structural parameters mediating RDH deformations. A Illustration of RDH structure and corresponding structural parameters, including torsion angle δ, χ, sugar pucker, and buckle. **B** Distributions of the torsion angle δ, χ, and sugar pucker related to the A/B-form of two strands in RDH compared between OL15/OL3 and CHARMM36 force field. **C** Distributions of inclination, buckle, and x-displacement are compared between OL15/OL3 and the CHARMM36 force field. **D**–**F** Correlation of δ with twist and slide, and the changes of δ with NaCl concentration from simulations using OL15/OL3 force field. **G**–**I** Same analysis as in (**G**–**I**) except using CHARMM36 force field. **J** Comparison of salt-induced twist change using OL15/OL3 and CHARMM36 with experimental results. **K** The PMF caused by the twist-slide coupling from simulations using the CHARMM 36 force field. The mean value over the corresponding MD trajectory block-averaged with a 100-ns interval, and standard errors are plotted as data points and error bars. Source data are provided as a Source Data file.

Except δ and χ, the correlations between twist, slide and other torsion angles or sugar pucker are rather weak (Supplementary Figs. 18–S24). See more details in Supplementary Note 13.

Ions can influence nucleic acid structures through both electrostatic screening and ion-specific binding. Our experimental results show that the twist responses of RDH to NaCl and KCl are broadly similar, with only modest differences. The overall similarity is consistent with the dominant role of electrostatic screening (Supplementary Fig. 25), whereas the observed deviations likely arise from ion-specific interactions. Correspondingly, our analysis shows that NaCl and KCl produce slightly different values of ∂D/∂sand ∂G/∂s, indicating that these ions differentially modulate the structural coupling between slide, duplex diameter, and major groove width. Notably, we found the major groove is highly salt-responsive in RNA, nearly rigid in DNA, and intermediate in RDH (Supplementary Fig. 26), consistent with the trend of f_G.

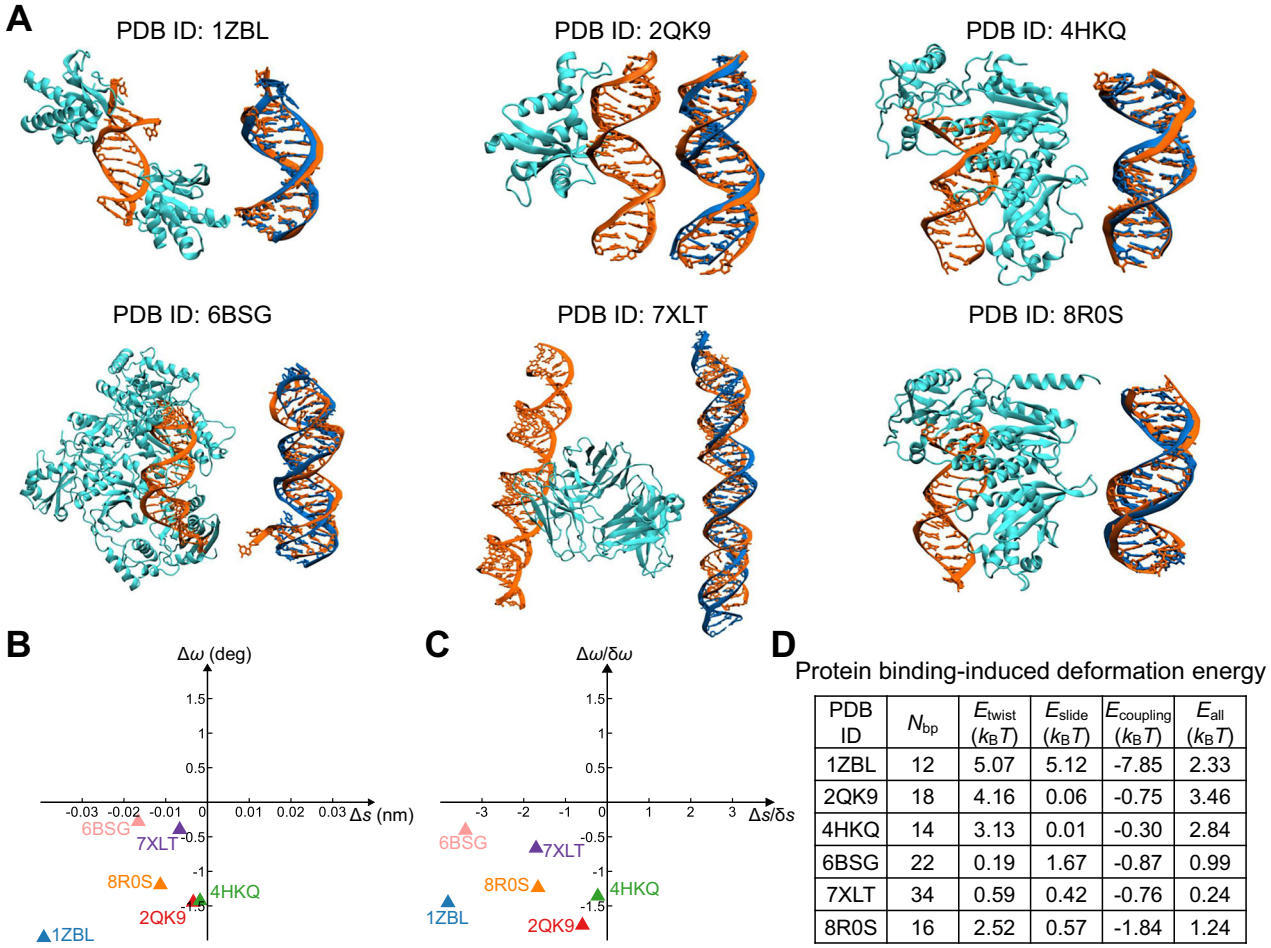
RDH deformations induced by protein binding

To examine whether protein-induced RDH deformations also occur along the direction dictated by twist-slide coupling, we performed MD simulations of six representative RDH–protein complexes (Fig. 6A) with complete structural information available. These complexes participate in diverse biological processes, including reverse transcription and RT-associated RNase H activity (4HKQ, 6BSG, and 8R0S), RNase H–mediated RDH recognition and cleavage (1ZBL and 2QK9), and protein-mediated RNA-DNA hybrid recognition and modulation (7XLT). Simulation results show that twist and slide remain strongly correlated in all complexes (Supplementary Table 6).

For each RDH–protein complex, we additionally performed MD simulations of the corresponding free RDH and calculated $\Delta\omega = \omega_{complex} - \omega_{free}$ and $\Delta s = s_{complex} - s_{free}$, where $\omega_{complex}$ and ω_{free} denote the twist values of the protein-bound and free RDH, respectively, and $s_{complex}$ and s_{free} denote the corresponding slide values. The scatter plot of $\Delta\omega$ versus Δs shows that protein-induced RDH deformations occur along the direction of twist-slide coupling (Fig. 6B).

Because different RDH sequences exhibit distinct intrinsic flexibilities, we further normalized $\Delta\omega$ and Δs by their thermal fluctuation amplitudes, $\delta\omega$ and δs, defined as the standard deviations of twist and slide fluctuations, respectively. The normalized results (Fig. 6C) and the corresponding deformation-energy analysis (Fig. 6D) show that

https://doi.org/10.1038/s41467-026-78514-9



D Protein binding-induced deformation energy

PDB ID	N_{bp}	E_{twist} (k_BT)	E_{slide} (k_BT)	$E_{coupling}$ (k_BT)	E_{all} (k_BT)
1ZBL	12	5.07	5.12	-7.85	2.33
2QK9	18	4.16	0.06	-0.75	3.46
4HKQ	14	3.13	0.01	-0.30	2.84
6BSG	22	0.19	1.67	-0.87	0.99
7XLT	34	0.59	0.42	-0.76	0.24
8R0S	16	2.52	0.57	-1.84	1.24

Fig. 6 | RDH deformations induced by protein binding. A Structures of representative RDH protein complexes (left in each pair) and comparisons between RDH conformations in the protein-bound (orange) and free states (blue, right in each pair). **B** Scatter plots showing protein binding-induced changes in RDH twist ($\Delta\omega$) and slide (Δs) per base pair. **C** Normalization by $\delta\omega$ and δs, which correspond to thermal fluctuation magnitudes. **D** Decomposition of RDH deformation energies related to twist and slide changes.

twist-slide coupling substantially reduces the conformational energy associated with protein-induced RDH deformations. These results suggest that twist-slide coupling provides an energetically favorable pathway for accommodating protein-induced RDH deformations.

Discussion

Measurements of twist changes by single-molecule magnetic tweezers (MT) provide a highly sensitive probe of nucleic-acid deformations. Although structural changes induced by salt, temperature, or mechanical perturbations often produce only minute twist variations at the level of individual base pairs, these changes accumulate over kilobase-length duplexes into experimentally detectable end rotations, enabling quantitative in situ measurements of otherwise difficult-to-resolve conformational responses. Combined with molecular dynamics simulations, this approach provides a powerful framework for dissecting deformation pathways in nucleic acids.

Using this strategy, we uncovered both distinctive mechanical behaviors and a common physical principle governing DNA, RNA, and RNA-DNA hybrid duplexes. Rather than behaving as a simple intermediate between DNA and RNA, RDHs exhibit markedly attenuated twist responses to salt and temperature, together with a nearly vanishing twist-stretch coupling. Despite these distinct mechanical properties, all three duplex types follow a universal deformation pathway: external perturbations first alter base-pair slide, which is subsequently converted into twist changes through twist-slide coupling constrained by the sugar-phosphate backbone. Importantly, this mechanism remains robust across molecular force fields. Although AMBER OL15/OL3 and CHARMM36 predict substantially different A-/B-form populations for the DNA strand in RDHs, they produce remarkably similar salt-induced twist responses and twist-slide coupling landscapes, indicating that twist-slide coupling is a common physical principle rather than a force-field-dependent feature.

The implications of this framework extend beyond environmentally induced deformations. RNA-DNA hybrids are transient but essential intermediates in numerous biological processes, including transcription[3,55,56], CRISPR-mediated genome editing[42,57], reverse transcription[58,59], and RNase H-mediated degradation[48,60,61], where proteins recognize, bind, and reshape RDHs. Our analysis further shows that protein-binding-induced RDH deformations occur along the direction dictated by twist-slide coupling, suggesting that this coupling represents an energetically preferred deformation pathway. Consequently, the intrinsic mechanics of RDHs are expected to influence protein recognition, binding affinity, and catalytic activity by governing the energetic cost of protein-induced conformational changes[62,63].

More broadly, the distinct yet predictable mechanical responses of DNA, RNA, and RDHs offer additional design routes for nucleic-acid nanotechnology[64,65]. Their different twist responses to ionic, thermal, and mechanical stimuli offer an additional design dimension beyond sequence alone, enabling programmable and dynamically

reconfigurable nucleic-acid nanostructures[66–68]. In particular, the attenuated environmental responses and nearly vanishing twist-stretch coupling of RDHs make them attractive as mechanically stable structural elements, whereas DNA and RNA can function as stimulus-responsive actuators. Together, our results establish twist-slide coupling as a unifying framework for understanding and engineering nucleic-acid deformations across different duplex chemistries, environmental perturbations, and protein interactions.

Methods

Magnetic-tweezers experiments

We built magnetic tweezers based on an inverted microscope (IX73, Olympus) by ourselves. The magnetic field was generated by a pair of separated NdFeB magnets with a fixed 1-mm gap above the flow cell. The vertical position of the magnets was precisely adjusted via a piezoelectric translation stage (PI, P-725.CD1), and magnet rotation was controlled by a high-precision stepper motor (PI, M126.PD1) to achieve accurate force loading and molecular torsion manipulation. A home-made collimated light-emitting diode (623 nm, 5 mW) served as the illumination source, which penetrated the gap between the dual magnets to uniformly illuminate magnetic beads inside the flow cell. Real-time bead image acquisition and three-dimensional coordinate tracking were performed by self-compiled LabVIEW software, enabling continuous monitoring of molecular extension changes during twisting and stretching. The temperature of the flow cell was stably controlled using a heating ring device (Objective Heater Controller: Bioptechs150803). The heating ring was tightly fitted around the objective to ensure uniform heat conduction. All salt-gradient and force-dependent torsion experiments were conducted at a constant temperature.

The labeled RDH construct was assembled via one-sided PCR (OSP) and in vitro annealing, with detailed protocols available in Supplementary Note 17 of SI and our prior work[69]. Briefly, we added bio-dUTP and dig-dUTP during separate PCR reactions to produce two DNA handles, which were enriched with abundant biotin and digoxigenin tags. The purified labeled handles were annealed with the target ssDNA and ssRNA fragment to form intact torsion-constructed RDH constructs. The products were purified to eliminate unannealed fragments. Gel electrophoresis was used to verify the purity and structural integrity of the final RDH samples.

The nucleic acid was labeled at one end with multiple digoxigenin groups and at the other end with multiple biotin groups. The digoxigenin end of nucleic acid was immobilized onto the inner bottom surface of the flow cell, which was coated with antidigoxigenin (Roche). Similarly, the biotin end of nucleic acid was tethered to a superparamagnetic microbead (DynabeadsTM MyOneTM Streptavidin C1). We used a pair of separated NdFeB magnets to apply constant forces to the magnetic bead, thereby stretching and rotating the NA. With the above experimental setup, we measured RDH twist changes induced by salt and force change using a procedure similar to our previous works.

Before formal torsion measurement, we fixed the applied force at 0.3 pN and rotated the RDH end using magnets. We only selected valid single-molecule RDH samples that could exhibit the maximum extension at the torsion-relaxed state. Samples with multiple tethered molecules, non-specific binding behaviors or abnormal rotation-extension curves were completely excluded to prevent experimental errors. During the rotation, we recorded the RDH extension to obtain the rotation-extension curve. This curve helped determine the torsionally relaxed point of RDH, which corresponds to the maximum extension. For each salt concentration of each ion type, we repeated the measurement using different RDH molecules in the flow cells. For data analysis, we used self-written MATLAB codes to process raw molecular extension and perform curve fitting and normalization to analyze RDH structural changes under different conditions. All experimental data were statistically analyzed, and the results were shown as mean ± standard errors to guarantee data reliability and repeatability.

Atomistic molecular dynamics simulations

The simulations started with a 25-bp RDH duplex in the sequence of 5′-CGACT CTACG GAAGG GCATC TGCGC-3′ for the DNA strand and its complementary one of 5′-GCGCA GAUGC CCUUC CGUAG AGUCG-3′ for the RNA strand. The initial structure was built using UCSF Chimaera 1.17[70] and placed in a rectangular box ($8 \times 8 \times 12.5$ nm), which was then filled with TIP3P water molecules. Counterions of Na^+, K^+, Rb^+ and a certain number of salt ions were added in the simulation box to match the concentrations in experiments. The details of each simulation system are listed in Supplementary Table 7.

All MD simulations were performed using the GROMACS 2018.4 software package[71]. We adopted the Joung-Cheatham model[72] for monovalent ions. The OL15 force filed[53] was used for DNA, and thw OL3 force field[54] is used to describe RNA. Periodic boundary conditions were applied in all three dimensions. A 1.0 nm cut-off was applied for van der Waals interactions and short-range electrostatic interactions. The Particle Mesh Ewald method was used for long-range electrostatic interactions. Covalent bonds involving hydrogen atoms were constrained using the LINCS algorithm. A 2 fs time step was used in conjunction with the leap-frog integrator. The temperature of the simulation system was coupled to 295 K using a V-rescale thermostat with a relaxation time $\tau = 0.1$ ps, and the pressure was kept at 1 atm using Parrinello-Rahman pressure-coupling with a relaxation time $\tau = 2.0$ ps and compressibility 4.5×10^{-5} bar^{-1}.

Each simulation system started with energy minimization using the steepest descent algorithm, which was followed by thermalization in the canonical (NVT) ensemble for 100 ps and equilibrated in the isobaric-isothermal (NPT) ensemble for 10 ns. In equilibration, DNA heavy atoms were constrained with a harmonic potential of 1000 kJ/(mol nm^2) to allow water and ions to relax around the duplex. After the equilibration, the constraints on heavy atoms were removed, and a production run of 600 ns was carried out for each simulation system. The last 500 ns of the production run was used for data analysis.

For simulations under different stretching forces, the constant force was applied to the mass-of-center of one terminal base pair along the end-to-end axis of the duplex, and another mass-of-center of the end base pair was fixed. For simulations under different temperatures, the temperature of the simulation system was coupled to desired values using the thermostat. 600 ns of production run was also conducted for each system, and the last 500 ns trajectory was collected for data analysis.

Helical parameters, including twist, slide, groove width, and diameter, were calculated from the production trajectories using Curves+ 3.0[73] or in-house codes. To minimize end-fraying effects, the terminal three base pairs at each end were excluded from the analysis. The reported mean values and uncertainties were obtained from block averaging and bootstrap analysis of the production trajectories. For DNA and RNA, we analyzed the MD simulations reported in our earlier work[25,28], performed with a similar methodology as the one used here for RDH.

Coarse-grained simulations

We performed coarse-grained MD simulations of a 60-bp duplex using oxNA model[49] for RDH. Different salt concentrations (0.1, 0.2, 0.3, 0.5, and 1 M) or temperatures (3, 13, 23, 33, and 43 °C) were applied in each simulation system. After obtaining the equilibrium configuration from Monte Carlo simulations of 10000 steps, we started the MD simulations for 10^9 steps. We use the Anderson-like thermostat with a time-step of 0.005 in reduced time unit[45].

Reporting summary
Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability
The data supporting the findings of this study are available from the corresponding authors upon request. Source data for the figures and Supplementary Figs. are provided as a Source Data file.

Code availability
The source code for simulations can be found via GitHub https://github.com/cityuBiophysics/NA_universal_twist_slide_coupling and Zenodo https://doi.org/10.5281/zenodo.22094455.

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Acknowledgments

We thank the Dai group members for discussions and advice.

Author contributions

L.D., X.Z., and F.T. conceived the idea; X.Z., W.W., C.Z., and F.T. performed the experiments; F.T., Q.Q, and L.D. performed the simulations and theoretical analysis.

Funding

We are grateful for the financial support from the National Natural Science Foundation of China (No. 12074294 to X.Z.; No. 12304254 to C.Z.; No. 22273080 to L.D.), China Postdoctoral Science Foundation [2025T003HB, 2025M773393], the Research Grants Council of Hong Kong (No. 11313322, 1130724, PDFS2425-1S09), and CityUHK Institute of Digital Medicine. Open Access made possible with partial support from the Open Access Publishing Fund of the City University of Hong Kong.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s41467-026-78514-9.

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Peer review information *Nature Communications* thanks Willem Vanderlinden and the other anonymous reviewer(s) for their contribution to the peer review of this work. A peer review file is available.

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